

Malware Classifications & Types of Malwares

What is Malware?

Malware (Malicious Software) is a type of software created with the intent to harm a computer system, network, or user data.

It can steal system resources, corrupt or delete data, give unauthorized access to attackers, or slow down and crash systems.

Malware Classifications

Malware is generally classified into **three major categories**:

1. Based on Propagation (How it Spreads):

This classification describes **how malware spreads** within or between systems.

- a. **Virus:** Attaches itself to a host file and spreads when the infected file is executed.
- b. **Worm:** Spreads automatically through networks without human interaction.
- c. **Trojan Horse:** Disguises itself as legitimate software but contains hidden malicious code.

2. Based on Action (Type of Attack or Damage):

This type of classification is based on the **behavior or intent** of the malware — what it does to the system.

- a. **Spyware:** Secretly monitors user activities and collects information.
- b. **Adware:** Displays unwanted advertisements on the user's device.
- c. **Ransomware:** Encrypts user data and demands a ransom (money) for decryption.
- d. **Rootkit:** Hides deep within the system and gives attackers administrative (root) control.

3. Based on Target (What It Attacks):

This defines **which part of the system or files** are targeted by the malware.

- a. **File Infectors:** Infect specific executable files.
- b. **Boot Sector Virus:** Targets the hard drive's boot sector.
- c. **Macro Virus:** Injects malicious code into macros in MS Word or Excel documents.
- d. **Network Malware:** Attacks network-based systems and communication protocols.

Types of Malwares

Type	Description	Effect / Example
Virus	A self-replicating program that attaches to a host file.	Corrupts files, slows system performance.
Worm	Automatically spreads through networks.	Consumes network bandwidth, causes denial of service.
Trojan Horse	Appears legitimate but performs hidden malicious actions.	Provides remote access to attackers.
Spyware	Secretly monitors user activities.	Steals sensitive data like passwords and keystrokes.
Adware	Displays unwanted advertisements.	Reduces system performance and user experience.
Ransomware	Encrypts user files and demands ransom for decryption.	Makes data inaccessible until payment is made.

Protection Against Malware

1. Install and regularly update reliable antivirus software.
2. Keep your operating system and all software up to date.
3. Avoid suspicious links, pop-ups, and unknown email attachments.
4. Use strong passwords and enable firewall protection.
5. Maintain regular data backups to prevent data loss.